Netaji Subhas University

1. Create a class `Car` with data members: brand, model, price. Add functions to display details.
2. Create a class `Time` to input hours and minutes and display them in proper format.
3. Create a class `Circle` with radius and function to calculate area and circumference.
4. Write a program to count the number of objects created using a static data member.
5. Demonstrate **Single Inheritance**
   a. Base: `Animal`
   b. Derived: `Dog`
6. Demonstrate **Multilevel Inheritance**

   - Class A → Class B → Class C

7. Write a program for **Hierarchical Inheritance**
8. Write a program for **Hybrid Inheritance**
9. Create a class `Complex` and overload `+` operator.
10. Create a class using **friend function**.
11. Write a program to demonstrate friend class.
12. Implement **Encapsulation** with private data members.
13. Demonstrate **Multiple Inheritance**.
14. Create an abstract class `Shape` and derive:

    - Rectangle
    - Circle

15. Write a program with function class name **Volume** using **Function Overloading** that computes the volume of a cube, a sphere, and a cuboid according to passed value.
    Formulae: Volume of a cube = (s * s * s)
    Volume of a sphere = (4 / 3 * 3.14 * r * r * r)
    Volume of a cube = (l * b * h)

16. Create a class that reads and writes objects to a file.
17. Write a C++ program using **exception handling** to manage bank transactions. Initialize the balance as **2000**, allow the user to **deposit** and **withdraw** money, and throw exceptions for **invalid deposit amounts** and **insufficient balance** during withdrawal.
18. Explain the role of **base class pointers** and **virtual functions with a neat C++ program**.

19. Write a program to:

    - Increment count in constructor
    - Decrement count in destructor
    - Display number of currently active objects

20. **Write a C++ program to calculate the surface area and volume of a cuboid using a constructor.**

Specifications:

- **Class Name:** `Cuboid`
- **Private Data Members:**
  - `length` – to store length of the cuboid
  - `breadth` – to store breadth of the cuboid
  - `height` – to store height of the cuboid
  - `surfaceArea` – to store surface area
  - `volume` – to store volume
- **Public Member Methods:**

  `1.Cuboid(float l, float b, float h)` : A **parameterized constructor** to initialize dimensions

  2. void compute() : To calculate surface area and volume of Cuboid.

  `3.void display()`: Displays the surface area and volume of the cuboid.

Formulae Used:

- Surface Area = `2(lb + bh + hl)`
- Volume = `l × b × h`

Also, write the main() method.